# DB Explorer — User Handbook

**Version:** 3.0.1
**Date:** April 2026
**Publisher:** Astro Adept AI Labs

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## 1. Overview

DB Explorer is a cross-platform database query tool and schema explorer built for developers, DBAs, and data analysts. It provides a clean graphical interface for connecting to multiple database systems, writing and executing SQL, exploring schemas, monitoring database health, and generating SQL using AI.

**Key capabilities:**

- Multi-database support (PostgreSQL, MySQL, MariaDB, SQL Server, Oracle, SQLite, DynamoDB)
- Multi-tab SQL editor with syntax highlighting and auto-complete
- Interactive schema browser
- Query result export (CSV, INSERT, UPDATE, DDL)
- Database health monitoring dashboard
- Execution plan viewer
- AI-powered SQL generation from natural language
- 13 built-in themes with live switching

## 2. System Requirements

| Component | Minimum | Recommended |
|---|---|---|
| OS | Windows 10, macOS 10.14, Ubuntu 18.04 | Windows 11, macOS 12+, Ubuntu 20.04+ |
| Java | Java 17 | Java 17 LTS |
| RAM | 512 MB | 2 GB |
| Disk | 100 MB | 500 MB |
| Display | 1024×768 | 1920×1080 |

## 3. Installation

### Option A — Pre-built Distribution (Recommended)

1. Download the latest release ZIP from the project repository.
2. Extract to any folder.
3. **JVM-bundled version:** run `db-explorer.exe` (Windows) or `./db-explorer` (Linux/macOS).
4. **JVM-independent version:** ensure Java 17+ is installed, then run:

```
java -jar db-explorer.jar
```

### Option B — Build from Source

**Prerequisites:** JDK 17+, Apache Maven 3.6+

```bash
git clone <repository-url>
cd db-explorer
mvn clean package
java -jar target/db-explorer-2.4.1.jar
```

# 4. Main Window Layout

```
┌──────────────────────────────────────────────────────────────┐
│  [+] [Run ▶] [Cancel ■] [New Tab] [Explain] [Health] [AI] [Theme ▾] │  ← Toolbar
├──────────────┬──────────────────────────────────────────────┤
│              │  Tab 1 ●  │  Tab 2 ●  │  + New Tab              │
│  Connections │──────────────────────────────────────────────│
│  Tree        │  SQL Editor                                   │
│              │                                               │
│  ▶ MyDB      │──────────────────────────────────────────────│
│    ▶ Tables  │  Results  │  Explain Plan                     │
│    ▶ Views   │                                               │
├──────────────┴──────────────────────────────────────────────┤
│  Status: Connected to MyDB  │  Heap: 256 MB / 1024 MB (25%)  [🗑]   │  ← Status bar
└──────────────────────────────────────────────────────────────┘
```

### Toolbar buttons:

| Button | Action |
|---|---|
| `+` | Add new database connection |
| `Run ▶` | Execute current SQL (green) |
| `Cancel ■` | Stop running query (red) |
| `New Tab` | Open a new query tab |
| `Explain` | Show execution plan for current query |
| `Health` | Open database health dashboard |
| `AI` | Open AI SQL Generator |
| `Theme ▾` | Switch UI theme instantly |
| `About` | Show application info |

### Status bar:

Shows active connection name, heap memory usage bar, and a garbage collection button (🗑) for manual memory management.

# 5. Managing Connections

## Create a Connection

1. Click the `+` toolbar button.

2. Fill in the connection form:

| Field | Description |
|---|---|
| Name | A friendly label for this connection |
| Type | Database type (PostgreSQL, MySQL, SQL Server, Oracle, SQLite, DynamoDB, etc.) |
| Host | Server hostname or IP address |
| Port | Port number (pre-filled with defaults) |
| Database | Database or schema name |
| Username | Login username |
| Password | Login password |

3. Click **Test Connection** to verify before saving.
4. Click **Save**.

## Connect / Disconnect

- **Connect:** Double-click a connection in the left panel, or right-click → **Connect**.
- **Disconnect:** Select the connection and click the **Disconnect** toolbar button, or right-click → **Disconnect**. A confirmation dialog prevents accidental disconnection.

## Edit / Delete

Right-click any connection in the left panel for **Edit** and **Delete** options.

## Supported Databases

| Database | Notes |
|---|---|
| PostgreSQL 9.0+ | Full schema, live stats, JSON/JSONB support |
| MySQL 5.5+ / MariaDB 10.0+ | Performance schema, stored procedures |
| Microsoft SQL Server 2012+ | DMVs, linked server browsing |
| Oracle 11g+ | PL/SQL objects, v$ views |
| SQLite 3.0+ | File-based, PRAGMA support |
| Amazon DynamoDB | NoSQL table browsing, PartiQL query execution, key schema display |
| Generic JDBC | Any JDBC-compatible database |

# 6. Writing and Running Queries

## Opening a Query Tab

- Click **New Tab** in the toolbar, or
- Right-click a connection → **Open Query Tab** to open a tab pre-bound to that connection.

### The SQL Editor

- Full syntax highlighting for SQL keywords, strings, numbers, and comments.
- Auto-complete triggers as you type — table names, column names, and SQL keywords.
- Each tab is independently bound to a connection — you can query multiple databases simultaneously.

### Running a Query

1. Type your SQL in the editor.
2. Press **Ctrl+Enter** or click the green **Run** button.
3. Results appear in the **Results** tab below the editor.

### Result Table

- Click any column header to sort ascending/descending/unsorted (cycles through states).
- Scroll horizontally for wide result sets.
- Large result sets are loaded lazily — scroll down to fetch more rows automatically.

### Cancelling a Query

Click the red **Cancel** button in the toolbar while a query is running.

## 7. Browsing the Schema

Expand a connected database in the left panel to browse:

- **Tables** — expand to see columns with data types
- **Views**
- **Stored Procedures / Functions**
- **Indexes and Constraints**

Double-click a table to open a query tab and view its data. Right-click items for context menu options.

Press **F5** to refresh the schema tree.

## 8. Exporting Data

After running a query, use the export buttons below the result table:

| Format | Description |
|---|---|
| CSV | Comma-separated values, compatible with Excel and most tools |
| INSERT | SQL INSERT statements for each row |
| UPDATE | SQL UPDATE statements |
| DDL | CREATE TABLE statement for the result schema |

Large exports are streamed in batches — no need to load all rows first.

# 9. Execution Plan

The execution plan shows how the database engine will execute your query — useful for identifying slow operations and missing indexes.

## How to Use

1. Write a SELECT query in the editor.
2. Click the **Explain** toolbar button.
3. The plan appears in the **Explain Plan** tab below the editor.
4. Long plan lines scroll horizontally — no text wrapping.

## Reading the Plan

| Term | Meaning |
|---|---|
| Seq Scan | Full table scan — consider adding an index |
| Index Scan | Efficient lookup via an index |
| Hash Join | Join using a hash table — good for large sets |
| Nested Loop | Join by iterating — best for small sets |
| `cost=X..Y` | Estimated startup and total cost (lower is better) |
| `rows=N` | Estimated row count |

## Database-Specific Behaviour

| Database | Method |
|---|---|
| PostgreSQL | `EXPLAIN ANALYZE` — shows actual runtime stats |
| MySQL / MariaDB | `EXPLAIN` — tabular format |
| SQL Server | `SET SHOWPLAN_TEXT ON` — text execution plan |
| Oracle | `EXPLAIN PLAN FOR` + `DBMS_XPLAN.DISPLAY()` |
| DynamoDB | Not supported |

> **Tip:** For PostgreSQL, use `EXPLAIN ANALYZE` directly in the SQL editor for actual execution times alongside estimates.

## 10. Database Health Dashboard

The health dashboard provides a live view of your database server's status.

### Opening

Click the **Health** toolbar button while connected to a database.

### Panels

| Panel | Information |
|---|---|
| Connection | Host, port, database name, driver version, ping latency |
| Server Stats | Uptime, active connections, cache hit ratio, database size |
| Active Queries | Currently running queries with duration and state |
| JVM Stats | Local heap usage, GC count, thread count |

Stats refresh automatically. Click **Refresh** for an instant update. Click the Health button again to close the panel.

## 11. AI SQL Generator

The AI SQL Generator converts natural language descriptions into SQL queries using your configured AI model.

### Prerequisites

Configure at least one AI profile first — see Section 13.

### Opening

Click the **AI** toolbar button. You must have an active database connection selected.

> The dialog opens at 85% of the main window size and adapts to your screen.

### Using the Generator

1. The **Database Context** bar at the top shows the connected database. Schema information is loaded automatically.
2. Select an **AI Profile** from the dropdown (top right).
3. Type your request in natural language in the **Describe Your Query** box.
   Example: *"Get all orders placed in the last 7 days with total value over 500"*
4. Click **Generate SQL**.
5. An animated progress bar shows while the AI is processing.
6. The generated SQL appears in the **Generated SQL** panel.
7. Click **Copy to Clipboard** or **Insert into Query Tab**.

### Tips

- The more specific your description, the better the result.
- The AI uses your actual schema (table and column names) for accurate output.
- Always review generated SQL before executing against production data.

## 12. Themes

DB Explorer includes 13 themes switchable at runtime with no restart required.

### How to Switch

Use the **Theme** dropdown in the toolbar. The change applies instantly to all open windows. Your selection is saved and restored on next launch.

### Available Themes

| Theme | Style | Best For |
|---|---|---|
| Flat Dark | Dark | Default — most environments |
| Flat Darcula | Warm dark | JetBrains IDE users |
| Flat Light | Light | Bright rooms, screenshots |
| Flat IntelliJ | Light | IntelliJ IDEA users |
| Ocean Blue | Dark blue | Long sessions, calm palette |
| Forest Green | Dark green | Earthy, relaxed feel |
| Sunset Purple | Dark purple | Vibrant, high contrast |
| Cherry Red | Dark + red accents | Bold, high energy |
| Amber Warm | Warm amber | Evening work |
| Arctic Frost | Cool light | Clean, airy feel |
| Rose Garden | Light + rose | Elegant light theme |
| System | OS native | Matches your OS |

| | | |
|---|---|---|
| Metal | Java default | Consistent cross-platform |

### Why Themes Matter

- **Dark themes** reduce eye strain in low-light environments.
- **Light themes** improve readability on bright displays and in printed screenshots.
- **High-contrast options** help users with visual sensitivities.

## 13. Settings Menu

Access via **Settings** in the menu bar ( `Alt+S` ).

### AI Configuration

**Settings** → **AI Configuration...** opens the AI profile manager.

| Field | Description |
|---|---|
| Name | A label for this profile |
| Provider | OpenAI, Claude, DeepSeek, Gemini, or Custom |
| Model | Exact model name (e.g. `gpt-4o` , `claude-sonnet-4-20250514` ) |
| API Key | Your provider API key — stored encrypted on disk |
| Base URL | Auto-filled per provider; change only for custom endpoints |
| Max Tokens | Maximum response length |
| Temperature | Creativity level (0.0 = deterministic, 1.0 = creative) |
| Enabled | Toggle this profile on/off |

**Steps:**

1. Click **Add New**.
2. Fill in the fields.
3. Click **Test Connection** — an animated progress bar shows while testing.
4. Click **Save**.
5. Check **Enable AI Assistant**.

Multiple profiles can be saved and switched from the AI Generator panel.

## 14. Help Menu

Access via **Help** in the menu bar ( `Alt+H` ).

| Menu Item | Opens |
|---|---|
| User Guide | Help dialog — Quick Start tab |
| DB Connection | Help dialog — DB Connection tab |
| AI Configuration | Help dialog — AI Configuration tab |
| DB Health Dashboard | Help dialog — DB Health tab |
| Execution Plan | Help dialog — Execution Plan tab |
| Themes | Help dialog — Themes tab |
| *(separator)* | |
| About DB Explorer | About dialog with version and build info |

# 15. Keyboard Shortcuts

| Shortcut | Action |
|---|---|
| `Ctrl+Enter` | Run current query |
| `Ctrl+Z` | Undo |
| `Ctrl+Y` | Redo |
| `Ctrl+N` | New query tab |
| `Ctrl+S` | Save current query |
| `Ctrl+F` | Find in results |
| `Ctrl+E` | Export results |
| `F5` | Refresh schema tree |
| `Alt+S` | Open Settings menu |
| `Alt+H` | Open Help menu |

# 16. Troubleshooting

## Application Won't Start

- Verify Java 17+: `java -version`
- Try increasing heap: `java -Xmx1024m -jar db-explorer.jar`
- Check console output for error messages.

## Connection Fails

- Confirm the database server is running and reachable.
- Check firewall rules for the port.
- Verify username and password.
- Use **Test Connection** in the connection dialog to diagnose.

### Out of Memory

- Increase heap: `java -Xmx2048m -jar db-explorer.jar`
- Reduce `query.max.rows` in `app.properties`.
- Use the 🗑 GC button in the status bar to free memory manually.

### Query is Slow

- Use **Explain Plan** to identify full table scans.
- Add indexes on filtered/joined columns.
- Use `LIMIT` / `FETCH FIRST N ROWS` to reduce result size.

### Execution Plan Shows Query Text Instead of Plan (SQL Server)

- Ensure the user has `SHOWPLAN` permission on the database.
- The plan requires a fresh connection — try disconnecting and reconnecting.

### AI Generator Returns an Error

- Verify the API key is correct in **Settings** → **AI Configuration...**.
- Use **Test Connection** to confirm the model is reachable.
- Check your API provider's quota and rate limits.

### Schema Tree is Empty After Connect

- Press **F5** to refresh.
- Confirm the user has metadata access privileges.
- For MySQL, ensure `information_schema` access is granted.

## 17. License

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